

Functional Traits Linked to Pathogen Prevalence in Wild Bee Communities

Supplementary Material

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Figure S1. Histogram showing distribution of bee size in *B. impatiens* workers. One clear outlier was removed from the *B. impatiens* intraspecific statistical analyses (furthest to the left).

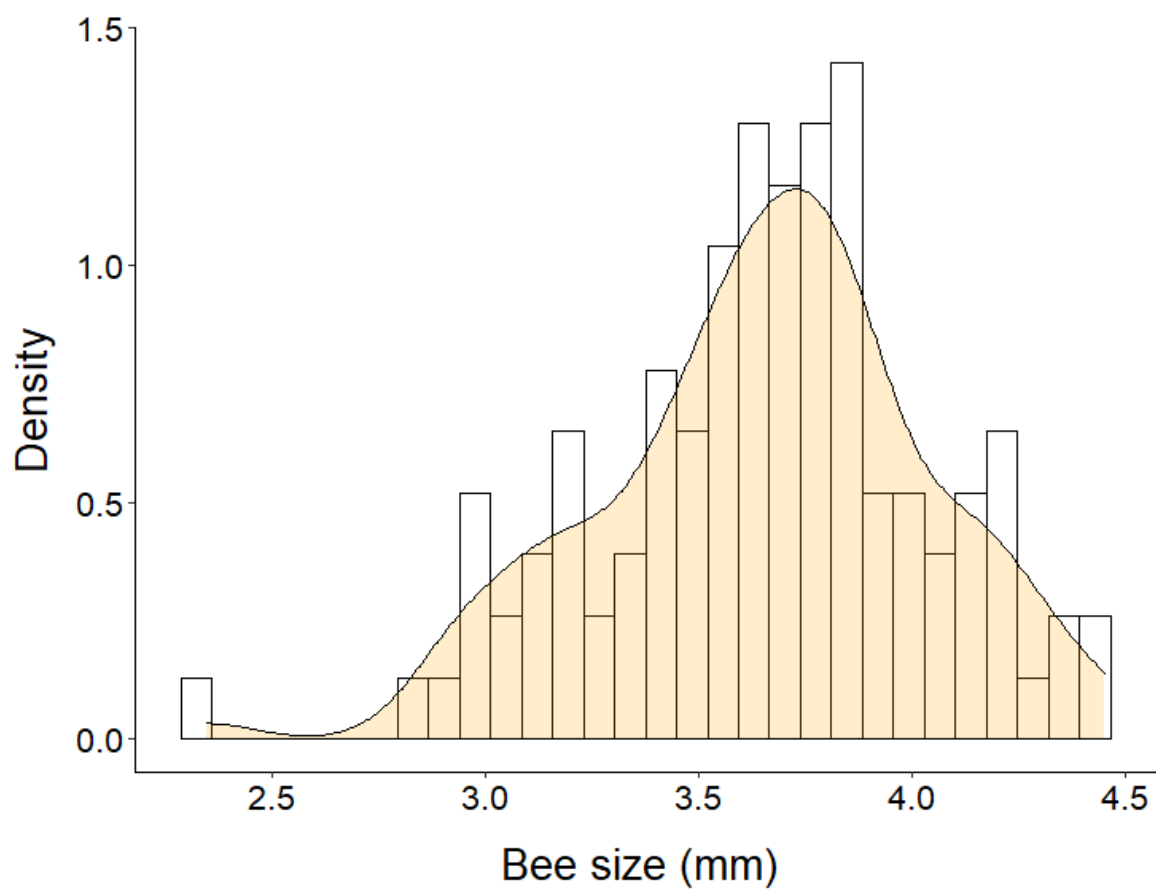


Figure S2. Relationship between individual *Bombus impatiens* worker size and likelihood of *Nosema ceranae* presence A) for all bees sampled and B) excluding the outlier (Figure S1).

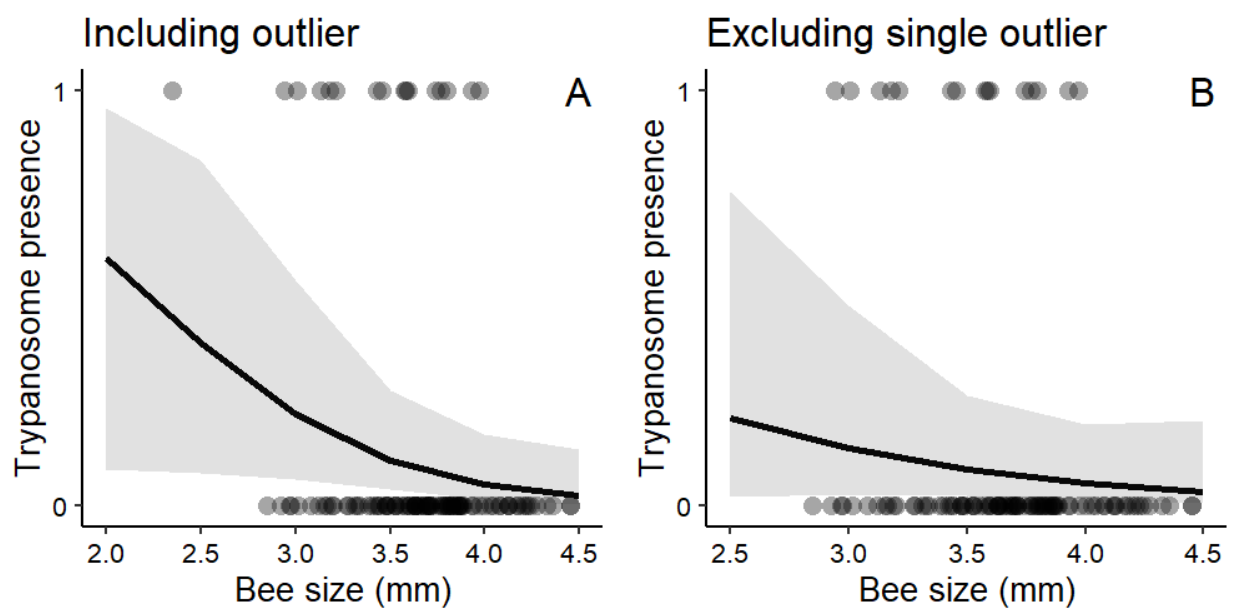


Figure S3. Relationship between bee size and trypanosome prevalence (mean $\pm$ SE) in the overall community (A) when *A. aurata* included and (B) when *A. aurata* excluded. Relationship between peak seasonal activity (in Julian days) and trypanosome prevalence (mean $\pm$ SE) in the overall community (A) when *A. aurata* included and (B) when *A. aurata* excluded. Data analyzed as continuous and plotted as such in Figure 4, presented here as proportion for ease of visualization.

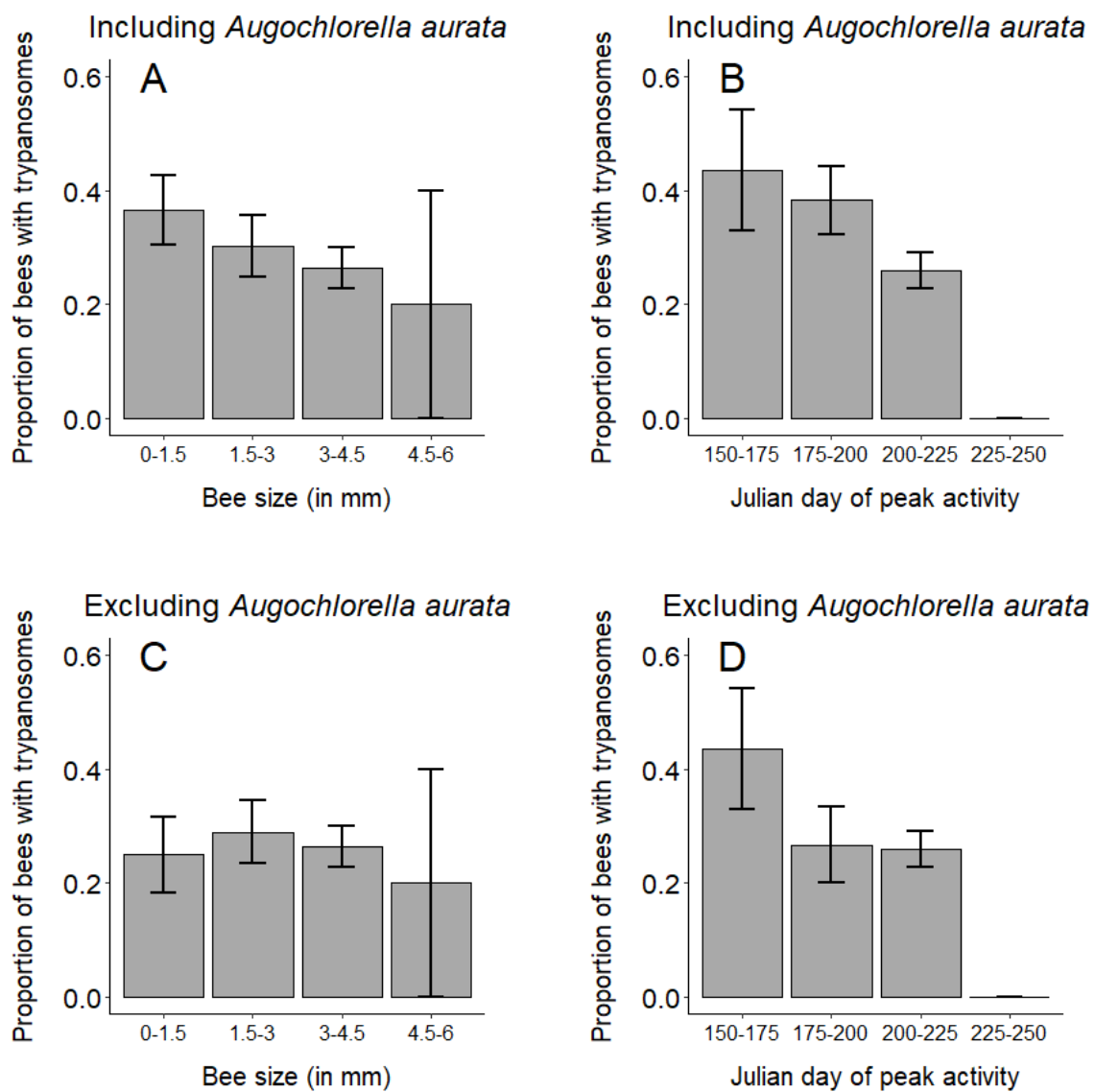


Figure S4. Histogram showing distribution of peak seasonal activity (based on Julian day) for the samples included in the functional trait analyses (A) and for the species in the NE of the US (overlapping region with the present study)¹.

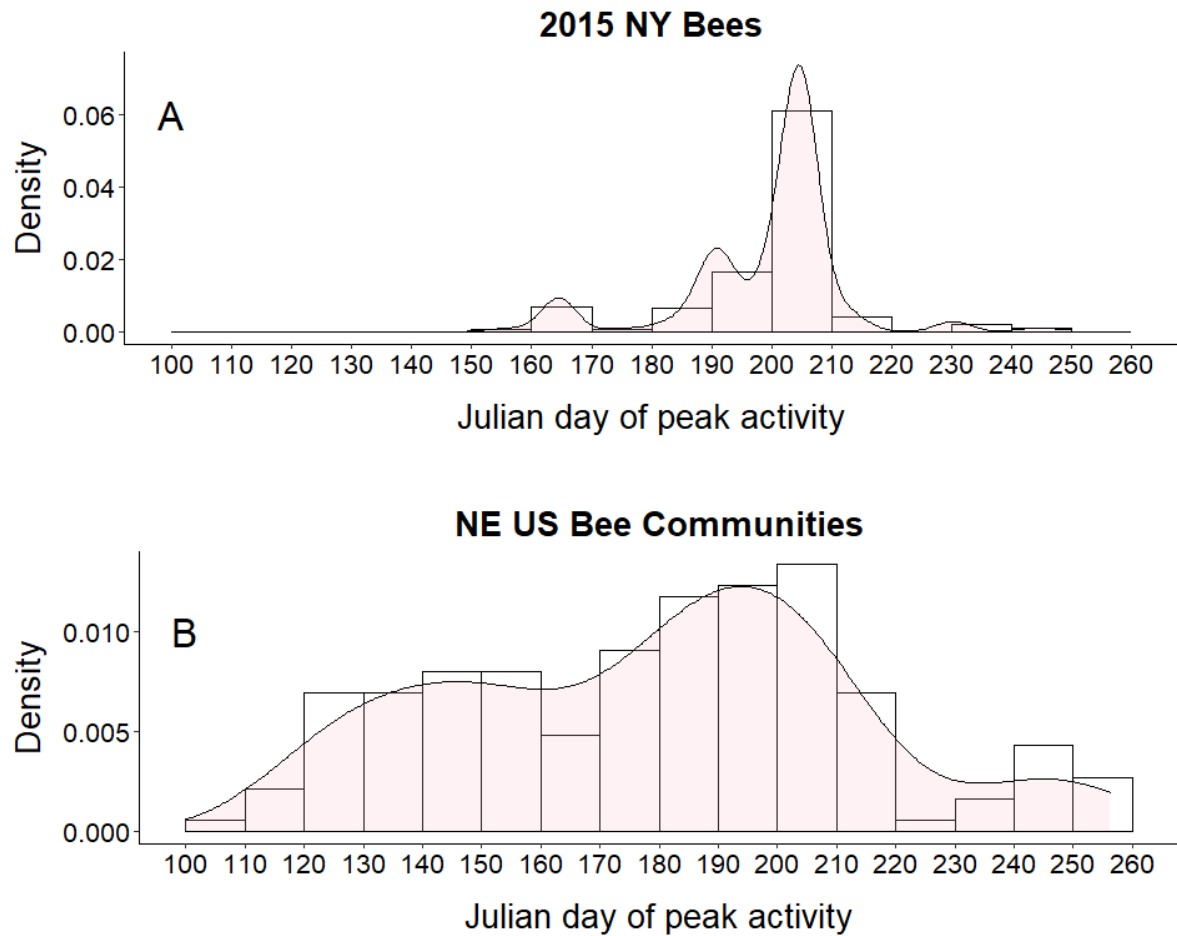


Table S1: Flower species present in wildflower planting and adjacent area where bees were collected.

In planting	Adjacent
<i>Agastache nepetoides</i>	<i>Convolvulus arvensis</i>
<i>Cirsium arvense</i>	<i>Daucus carota</i>
<i>Coreopsis lanceolata</i>	<i>Hypericum perforatum</i>
<i>Daucus carota</i>	<i>Leucanthemum vulgare</i>
<i>Erigeron annuus</i>	<i>Lotus corniculatus</i>
<i>Lobelia siphilitica</i>	<i>Malva moschata</i>
<i>Medicago sativa</i>	<i>Potentilla fruticosa</i>
<i>Penstemon digitalis</i>	<i>Plantago lanceolata</i>
<i>Silphium perfoliatum</i>	<i>Rubus fruticosus</i>
<i>Solidago juncea</i>	<i>Symphyotrichum novi-belgii</i>
<i>Trifolium pratens</i>	
<i>Trifolium repens</i>	
<i>Veronicastrum virginicum</i>	

Table S2: Plant families visited by each bee species across the 11 sampled sites.

Bee species	Plant Family	<i>n</i>
<i>Agapostemon virescens</i>	Asteraceae	4
	Plantaginaceae	7
<i>Andrena simplex</i>	Asteraceae	1
<i>Andrena spp.</i>	Asteraceae	1
<i>Andrena wilkella</i>	Fabaceae	2
<i>Anthophora terminalis</i>	Plantaginaceae	22
	NA	1
<i>Apis mellifera</i>	Asteraceae	33
<i>Augochlora pura</i>	Plantaginaceae	2
<i>Augochlorella aurata</i>	Asteraceae	9
	Convolvulaceae	6
	Fabaceae	2
	Plantaginaceae	8
<i>Bombus bimaculatus</i>	Asteraceae	3
	Fabaceae	4
	Plantaginaceae	16
<i>Bombus fervidus</i>	Asteraceae	1
	Fabaceae	1
	Plantaginaceae	1
<i>Bombus griseocollis</i>	Asteraceae	4
	Plantaginaceae	5
<i>Bombus impatiens</i>	Apiaceae	2
	Asteraceae	99
	Campanulaceae	3
	Fabaceae	1
	Lamiaceae	5
	Plantaginaceae	12
<i>Bombus perplexus</i>	Asteraceae	1
<i>Bombus ternarius</i>	Asteraceae	3
	Plantaginaceae	2
<i>Bombus vagans</i>	Asteraceae	3
	Fabaceae	1
	Plantaginaceae	6
<i>Calliopsis andreniformis</i>	Asteraceae	1
<i>Calliopsis nebraskensis</i>	Asteraceae	1
<i>Ceratina calcarata</i>	Asteraceae	5
	Campanulaceae	3
	Malvaceae	1
	Plantaginaceae	3
	Rosaceae	1
	NA	2
<i>Ceratina dupla</i>	Plantaginaceae	1

	Rosaceae	1
<i>Ceratina mikmaqi</i>	Asteraceae	8
	Fabaceae	1
	Hypericaceae	1
	Plantaginaceae	8
<i>Ceratina spp.</i>	Asteraceae	2
	Campanulaceae	1
	Plantaginaceae	3
<i>Coelioxys rufitarsis</i>	Asteraceae	1
<i>Colletes kincaidii</i>	Asteraceae	1
<i>Halictus confusus</i>	Asteraceae	13
	Convolvulaceae	1
	Plantaginaceae	2
	Rosaceae	1
<i>Halictus ligatus</i>	Asteraceae	32
	Convolvulaceae	8
<i>Halictus rubicundis</i>	Fabaceae	1
<i>Heriades carinata</i>	Asteraceae	1
<i>Hoplitis producta</i>	Plantaginaceae	1
<i>Hoplitis spp.</i>	Asteraceae	1
<i>Hylaeus affinis</i>	Asteraceae	6
	Convolvulaceae	2
	Plantaginaceae	3
<i>Hylaeus annulatus</i>	Asteraceae	1
	Malvaceae	1
	Plantaginaceae	1
<i>Hylaeus modestus</i>	Asteraceae	1
	Campanulaceae	9
	Plantaginaceae	4
<i>Hylaeus spp.</i>	Asteraceae	3
<i>Lasioglossum coriaceum</i>	Plantaginaceae	3
<i>Lasioglossum hitchensi</i>	Plantaginaceae	1
<i>Lasioglossum imitatum</i>	Asteraceae	2
<i>Lasioglossum inconditum</i>	NA	1
<i>Lasioglossum lineatulum</i>	Asteraceae	1
	Plantaginaceae	2
<i>Lasioglossum perpunctatum</i>	Asteraceae	1
<i>Lasioglossum pilosum</i>	Asteraceae	4
	Plantaginaceae	7
<i>Lasioglossum spp.</i>	Asteraceae	10
	Convolvulaceae	1
	Hypericaceae	4
	Plantaginaceae	10
<i>Lasioglossum versatum</i>	Plantaginaceae	3
<i>Lasioglossum viridatum</i>	Asteraceae	7

	Convolvulaceae	4
	Fabaceae	1
	Plantaginaceae	31
	Rosaceae	2
	NA	1
<i>Lasioglossum weemsi</i>	Campanulaceae	1
	Asteraceae	3
<i>Lasioglossum zephyrum</i>	Hypericaceae	3
	Plantaginaceae	1
<i>Megachile spp.</i>	Asteraceae	1
	Plantaginaceae	4
<i>Melissodes agilis</i>	Asteraceae	5
<i>Melissodes bimaculata</i>	Asteraceae	4
<i>Melissodes desponsa</i>	Asteraceae	6
<i>Melissodes druriella</i>	Asteraceae	3
<i>Melissodes spp.</i>	Asteraceae	37
	NA	1
<i>Melissodes subillata</i>	Asteraceae	3
<i>Melissodes trinodis</i>	Asteraceae	29
<i>Osmia atriventris</i>	Plantaginaceae	1
<i>Osmia bucephala</i>	Plantaginaceae	4
<i>Peponapis pruinosa</i>	Asteraceae	1
<i>Xylocopa virginica</i>	Asteraceae	2

Table S3. Molecular conditions used for pathogen screening (PCR).

Primers and source	Thermal Cycling			Amplicon size (bp)
	Denaturin g Min Temp	Replication Sec Temp	Elongation Min Temp	
<i>Apidae^A (host) and Trypanosomes^{T 2}</i>	2 94	35x 30 94 30 61 45 68	5 68	130 ^A 420 ^T
ApidaeF(5-3): AGATGGGGGCATTTCGTATTG				
ApidaeR(5-3): ATCTGATCGCCTTCGAACCT				
SEF(5-3): CTTTGGTTCGGTGGAGTGAT				
SER(5-3): GGACGTAATCGGCACAGTTT				
<i>Neogregarines²</i>	2 94	35x 30 94 30 60.7 45 68	3 68	260
NeoF (5-3): CCAGCATGGAATAACATGTAAGG				
NeoR(5-3): GACAGCTTCCAATCTCTAGTCG				
<i>Nosema apis^A, N ceranae^{Nc 3}</i>	4 95	35x 60 95 60 60 60 68	5 68	297 ^{Na} 662 ^{Nc}
NosaRNAPol-F2 (5-3): AGCAAGAGACGTTTCTGGTACCTCA				
NosaRNAPol-R2 (5-3): CCTTCACGACCACCCATGGCA				
NoscRNAPol-F2 (5-3): TGGGTCCCTAAACCTGGTGGTTT				
NoscRNAPol-R2 (5-3): TCACATGACCTGGTGCTCCTTCT				
<i>Nosema bombi⁴</i>	4 95	35x 60 95 60 50 60 68	4 68	323
Nbombi-SSU-Jf1 (5-3): CCA TGC ATG TTT TTG AAG ATT ATT AT				
Nbombi-SSU-Jr1 (5-3): CAT ATA TTT TTA AAA TAT GAA ACA ATA A				

Table S4: Summary of bee species and functional traits. Sample size refers to bees for which we had complete functional trait information, including sex ($n = 289$). Peak activity and Activity range in Julian days.

Family	Bee species	Nesting location	Sociality	Peak activity	Activity range	Average ♀ size (mm)	♀ n	Average ♂ size (mm)	♂ n
Halictidae	<i>Agapostemon virescens</i>	Below-ground	Social	212	98	2.27	8		
Andrenidae	<i>Andrena simplex</i>	Below-ground	Solitary	250	29	2.00	1		
Andrenidae	<i>Andrena sp</i>	Below-ground	Solitary			1.82			
Andrenidae	<i>Andrena wilkella</i>	Below-ground	Solitary			2.35			
Apidae	<i>Anthophora terminalis</i>	Above-ground	Solitary	201	73	3.12	13	2.55	1
Apidae	<i>Apis mellifera</i>	Above-ground	Social			3.03			
Halictidae	<i>Augochlora pura</i>	Above-ground	Solitary	205	116	1.74	2		
Halictidae	<i>Augochlorella aurata</i>	Below-ground	Social	191	99	1.45	20	1.26	3
Apidae	<i>Bombus bimaculatus</i>	Below-ground	Social	165	84	3.83	18	3.78	2
Apidae	<i>Bombus fervidus</i>	Below-ground	Social	206	117	3.58	1	3.98	2
Apidae	<i>Bombus griseocollis</i>	Below-ground	Social	201	96	4.22	4	5.00	3
Apidae	<i>Bombus impatiens</i>	Below-ground	Social	205	129	3.65	107		
Apidae	<i>Bombus perplexus</i>	Below-ground	Social	182	95				
Apidae	<i>Bombus ternarius</i>	Below-ground	Social	206	94	2.91	2	3.43	3
Apidae	<i>Bombus vagans</i>	Below-ground	Social	201	101	3.39	10		
Andrenidae	<i>Calliopsis andreniformis</i>	Below-ground	Solitary	188	51			1.22	1
Andrenidae	<i>Calliopsis nebraskensis</i>	Below-ground	Solitary					1.12	
Apidae	<i>Ceratina calcarata</i>	Above-ground	Social	183	116	1.42	5		
Apidae	<i>Ceratina dupla</i>	Above-ground	Social			1.08			
Apidae	<i>Ceratina mikmaqi</i>	Above-ground	Solitary			1.53		1.17	
Apidae	<i>Ceratina sp</i>	Above-ground				1.35			
Megachilidae	<i>Coelioxys rufitarsis</i>	Below-ground		211	78				
Colletidae	<i>Colletes kincaidii</i>	Below-ground	Social	213	36	2.57	1		
Halictidae	<i>Halictus confusus</i>	Below-ground	Social	189	99	1.36	4	1.35	8
Halictidae	<i>Halictus ligatus</i>	Below-ground	Social	205	95	1.61	28		
Halictidae	<i>Halictus rubicundis</i>	Below-ground	Social						
Megachilidae	<i>Heriades carinata</i>	Above-ground	Solitary			1.52			
Megachilidae	<i>Hoplitis producta</i>	Above-ground	Solitary	175	45	1.66	1		
Megachilidae	<i>Hoplitis sp</i>	Above-ground	Solitary						
Colletidae	<i>Hylaeus affinis</i>	Above-ground	Solitary	197	71	1.17	6	1.18	1
Colletidae	<i>Hylaeus annulatus</i>	Above-ground	Solitary			1.09			
Colletidae	<i>Hylaeus modestus</i>	Above-ground	Solitary	194	75	1.08	6	1.16	1
Colletidae	<i>Hylaeus sp</i>	Above-ground	Solitary						
Halictidae	<i>Lasioglossum coriaceum</i>	Below-ground	Social	202	112	2.02	1		
Halictidae	<i>Lasioglossum hitchensi</i>	Below-ground	Social						
Halictidae	<i>Lasioglossum imitatum</i>	Below-ground	Social	194	106				
Halictidae	<i>Lasioglossum inconditum</i>	Below-ground	Solitary			1.25			
Halictidae	<i>Lasioglossum lineatulum</i>	Below-ground	Social	188	91	1.29	1		
Halictidae	<i>Lasioglossum perpunctatum</i>	Below-ground	Social	176	89	1.31	1		
Halictidae	<i>Lasioglossum pilosum</i>	Below-ground	Social	191	102	1.28	5		
Halictidae	<i>Lasioglossum sp</i>	Below-ground				1.33		1.33	
Halictidae	<i>Lasioglossum versatum</i>	Below-ground	Social	193	100	1.20	2		
Halictidae	<i>Lasioglossum viridatum</i>	Below-ground	Social			1.11		1.74	
Halictidae	<i>Lasioglossum weemsi</i>	Below-ground	Social	182	103				
Halictidae	<i>Lasioglossum zephyrum</i>	Below-ground	Social	191	136	1.08	4		
Megachilidae	<i>Megachile sp</i>		Solitary			4.00		2.92	
Apidae	<i>Melissodes agilis</i>	Below-ground	Solitary					2.44	
Apidae	<i>Melissodes bimaculata</i>	Below-ground	Solitary	214	42			2.81	2
Apidae	<i>Melissodes desponsa</i>	Below-ground	Solitary	230	41	2.59	1	2.88	5
Apidae	<i>Melissodes dhuriella</i>	Below-ground	Solitary	244	37	2.53	2		
Apidae	<i>Melissodes sp</i>	Below-ground	Solitary			2.67		2.61	
Apidae	<i>Melissodes subillata</i>	Below-ground	Solitary					2.14	
Apidae	<i>Melissodes trinodis</i>	Below-ground	Solitary			2.86		2.52	
Megachilidae	<i>Osmia atriventris</i>	Above-ground	Solitary	157	65	2.23	1		
Megachilidae	<i>Osmia bucephala</i>	Above-ground	Solitary	155	60	4.00	1		
Apidae	<i>Peponapis pruinosa</i>	Below-ground	Solitary	215	49	2.97	1		
Apidae	<i>Xylocopa virginica</i>	Above-ground	Social	176	116				

Table S5: Post-hoc comparison of differences between genera. Comparisons that resulted in P

values ≥ 0.9 are not shown. The estimate correspond to the Estimated Marginal Means.

Pathogen	Contrast	Estimate	SE	z ratio	P value
Trypanosoma	<i>Augochlorella</i> - <i>Lasioglossum</i>	2.52	0.60	4.16	0.005
Trypanosoma	<i>Augochlorella</i> - <i>Bombus</i>	1.91	0.50	3.85	0.017
Trypanosoma	<i>Augochlorella</i> - <i>Apis</i>	3.17	0.88	3.61	0.040
Trypanosoma	<i>Halictus</i> - <i>Lasioglossum</i>	1.69	0.54	3.15	0.158
Trypanosoma	<i>Agapostemon</i> - <i>Apis</i>	3.34	1.07	3.12	0.174
Trypanosoma	<i>Agapostemon</i> - <i>Lasioglossum</i>	2.68	0.86	3.10	0.180
Trypanosoma	<i>Lasioglossum</i> - <i>Melissodes</i>	-1.61	0.52	-3.07	0.195
Trypanosoma	<i>Apis</i> - <i>Halictus</i>	-2.35	0.83	-2.84	0.328
Trypanosoma	<i>Apis</i> - <i>Melissodes</i>	-2.27	0.82	-2.76	0.386
Trypanosoma	<i>Bombus</i> - <i>Halictus</i>	-1.09	0.41	-2.68	0.443
Trypanosoma	<i>Bombus</i> - <i>Melissodes</i>	-1.01	0.38	-2.66	0.459
Trypanosoma	<i>Agapostemon</i> - <i>Bombus</i>	2.08	0.79	2.63	0.479
Trypanosoma	<i>Augochlorella</i> - <i>Ceratina</i>	1.68	0.67	2.50	0.585
Trypanosoma	<i>Augochlorella</i> - <i>Hylaeus</i>	1.66	0.73	2.27	0.754
Trypanosoma	<i>Agapostemon</i> - <i>Ceratina</i>	1.84	0.89	2.07	0.873
<i>N. ceranae</i>	<i>Anthophora</i> - <i>Bombus</i>	2.22	0.64	3.47	0.063
<i>N. ceranae</i>	<i>Bombus</i> - <i>Melissodes</i>	-1.49	0.47	-3.15	0.161
<i>N. ceranae</i>	<i>Bombus</i> - <i>Megachile</i>	-3.84	1.26	-3.06	0.200
<i>N. ceranae</i>	<i>Augochlorella</i> - <i>Megachile</i>	-3.97	1.43	-2.78	0.371
<i>N. ceranae</i>	<i>Anthophora</i> - <i>Lasioglossum</i>	1.91	0.7	2.75	0.394

			0		
<i>N. ceranae</i>	<i>Lasioglossum - Megachile</i>	-3.53	1.2 9	-2.74	0.397
<i>N. ceranae</i>	<i>Bombus - Halictus</i>	-1.37	0.5 1	-2.67	0.452
<i>N. ceranae</i>	<i>Anthophora - Augochlorella</i>	2.35	0.9 4	2.50	0.582
<i>N. ceranae</i>	<i>Apis - Bombus</i>	1.30	0.5 7	2.28	0.748
<i>N. ceranae</i>	<i>Ceratina - Megachile</i>	-2.93	1.3 5	-2.17	0.822
<i>N. ceranae</i>	<i>Lasioglossum - Melissodes</i>	-1.18	0.5 7	-2.08	0.866

Table S6: Summary table of functional traits and pathogen prevalence models. *P* values < 0.05 are in **bold**.

Pathogen	Functional trait	Estimate	SE	<i>z</i>	<i>P</i>
Trypanosomes	Bee size	-0.45	0.16	-2.72	0.006
Trypanosomes	Peak seasonal activity	-0.42	0.15	-2.72	0.007
Trypanosomes	Seasonal activity range	0.04	0.19	0.19	0.853
Trypanosomes	Nesting location	0.81	0.48	1.71	0.088
Trypanosomes	Sex	-0.31	0.49	-0.63	0.532
Trypanosomes	Flower family	0.23	0.32	0.71	0.479
Trypanosomes ≠ <i>A. aurata</i>	Bee size	-0.25	0.17	-1.46	0.144
Trypanosomes ≠ <i>A. aurata</i>	Peak seasonal activity	-0.40	0.16	-2.49	0.013
Trypanosomes ≠ <i>A. aurata</i>	Seasonal activity range	-0.01	0.20	-0.05	0.959
Trypanosomes ≠ <i>A. aurata</i>	Nesting location	0.50	0.50	1.00	0.317
Trypanosomes ≠ <i>A. aurata</i>	Sex	-0.44	0.54	-0.81	0.419
Trypanosomes ≠ <i>A. aurata</i>	Flower family	-0.08	0.36	-0.22	0.830
<i>Nosema ceranae</i>	Bee size	-0.12	0.27	-0.45	0.652
<i>Nosema ceranae</i>	Peak seasonal activity	0.31	0.26	1.19	0.235
<i>Nosema ceranae</i>	Seasonal activity range	0.03	0.29	0.09	0.931
<i>Nosema ceranae</i>	Nesting location	-1.24	0.65	-1.89	0.059
<i>Nosema ceranae</i>	Sex	-0.98	0.87	-1.12	0.263
<i>Nosema ceranae</i>	Flower family	0.07	0.44	0.17	0.869
Neogregarines	Bee size	0.54	0.32	1.68	0.093
Neogregarines	Peak seasonal activity	-0.12	0.24	-0.50	0.620
Neogregarines	Seasonal activity range	-0.36	0.29	-1.27	0.205
Neogregarines	Nesting location	18.57	332.45	0.06	0.956
Neogregarines	Sex	-19.28	458.25	-0.04	0.966
Neogregarines	Flower family	-0.87	0.65	-1.34	0.182

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- 3 Gisder, S. & Genersch, E. Molecular differentiation of *Nosema apis* and *Nosema ceranae* based on species-specific sequence differences in a protein coding gene. *J. Invertebr. Pathol.* **113**, 1-6 (2013).
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